Reg.No.						

Bansilal Ramnath Agarwal Charitable Trust's

VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE – 411037.

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Examination: ESE Format

Year: F. Y. B. Tech **Branch**: Common to All

Subject: Linear Algebra and Differential Equations **Subject Code: HS1071**

Max. Marks: 60 **Total Pages of Question Paper:**

Day & Date: Thursday, 5th June 2025 **Time**:

Instructions to Candidate

1. All questions are compulsory.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right indicate full marks.

Q. N.	CO No	BT* No	Question	Max marks
Q. 1.	1	1,2, 3,4,5	**Attempt the following**	5
			System of Equations	
Q. 2.	2		**Attempt Any ONE of the following**	5
a)		3,4, 5,6	Linear Combination/ Span/ Spanning Set/Linear Dependence-Independence /Basis/Dimension	
b)		2,3 4,5		
Q. 3.	3		**Attempt the following**	5
		1,2, 3,4,5	Inner Product / Norm /Distance /Orthogonality	03/02
		1,2, 3,4,5		02/03
Q. 4(A)	4		**Attempt Any TWO of the following**	12
a)		2,3, 4,5	Linear Transformation Definition / Kernel/ Range/One-One, Onto transformation/Regular transformation/2D Geometric Transformations.	
b)		1,2,3, 4,5,6		
c)		2,3, 4,5		
(B)			**Attempt the following**	03
		1,2 3,4	Linear Transformation Definition / Regular transformation/ 2D Geometric Transformations.	
Q. 5(A)	5		**Attempt Any TWO of the following**	12
a)		2,3, 4,5	Eigen value, Eigen vector Definition / Properties of Eigen values, Eigen vectors/Diagonalization/Quadratic Form	
b)		2,3, 4,5		
c)		1,2, 3,4,5		
(B)			**Attempt the following**	03
		1,2 3,4,5	Eigen value, Eigen vector Definition / Characteristic Equation	
Q. 6 (A)	6		**Attempt Any TWO of the following**	12
a)		2,3, 4,5	Solution with initial conditions/ Method of variation of parameters/Method of undetermined coefficients	
b)		2,3, 4,5		
c)		2,3, 4,5,6		
(B)			**Attempt the following**	03
		1,2 3	Complementary solution	

CO Statements:

CO1: The student will be able to set up, solve and analyze linear systems of equations

CO2: The student will be able to understand the concepts of vector spaces, subspaces, spanning set, basis, linear dependence and independence.

CO3: The student will be able to apply knowledge of inner product spaces to compute length of a vector, angle, distance between two vectors, to compute orthogonal basis using Gram-Schmidt process.

CO4: The student will be able to demonstrate linear transformations geometrically and find basis and kernel of linear transformation.

CO5: The student will be able to compute Eigen values and Eigen vectors and apply it for diagonalization of matrices.

CO6: The student will be able to solve and interpret solution of linear differential equations representing models various physical processes.

***Blooms Taxonomy (BT) Level No:**

- 1. Remembering; 2. Understanding; 3. Applying; 4. Analyzing; 5. Evaluating; 6. Creating